EET202

HINES & TRANSFORMERS

MODULE: 5

Lecture:1 –

Prepared By,

ADARSH SR
Asst. Professor
EEE Department
SCET Kodakara

Syllabus

ng of copper –ratingof autotransformers.

er – construction- difference between power transformer and
–Different connections of 3-phase transformers. Y-Y, Δ-Δ,Y-Δ,
pings – Yy0, Dd0, Yd1, Yd11, Dy1, Dy11.Parallel operation of
s.

mer – stabilization by tertiary winding. Tap changing transformers
on load tap changing, dry type transformers.

e Phase Transformer

transmit electric power, a three-phase system is

ystem if we want to step-up (raise) or step-down
ges, 3-phase transformers are used.

rs can be built in two ways.

-phase transformers (3-phase transformer bank)

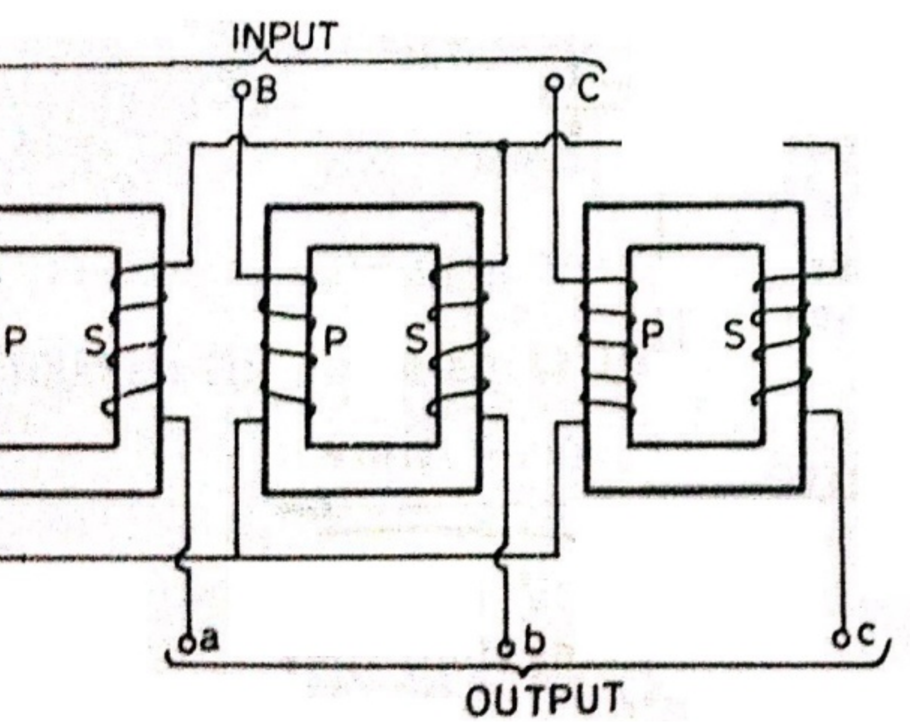
nsformer as a single unit

Construction

nits of single phase transformers are used to form

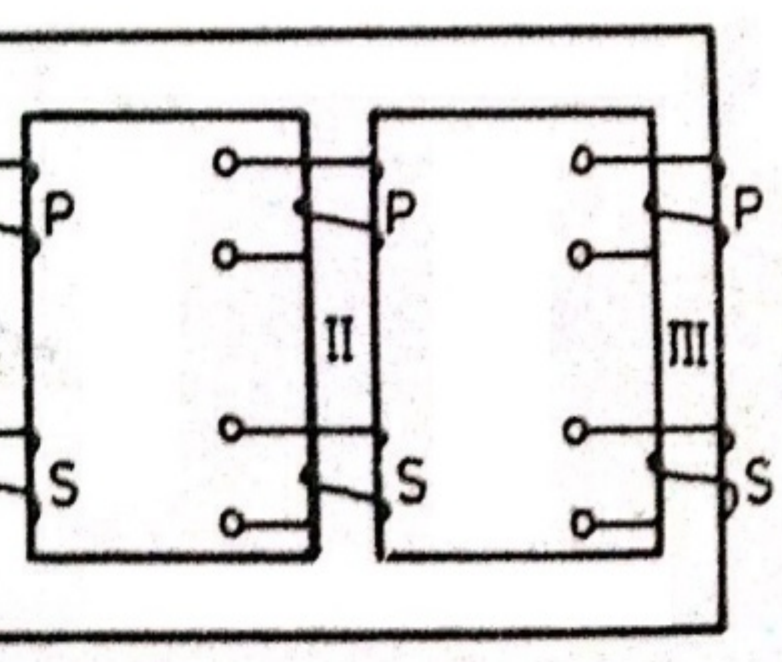
nsformer.

for each phase.



Construction

constructed by having 3 primary and 3
ngs on a common frame.
type or of shell type construction.



Construction

nstruction, 3-limbed core is used, each limb for
ding.

nstruction, there will be a 5-limbed core and the
e having the windings.

mary and secondary windings are placed on the

w-voltage (LV) windings are placed adjacent to
igh-voltage (HV) windings are placed over LV

ulation is placed between the core and LV
so in between LV and HV windings.

Power vs. Distribution Transformers

former	Distribution Transformer
sion network of nd higher	It is used in the distribution network of lower voltages 11KV and lower
200 MVA	Used for rating below 200 MVA
compared of ers	Smaller in size
stations and ns	Used in distribution stations, also for industrial and domestic purposes
l load	Operated at load less than full load as load cycle fluctuates
l as the ratio of wer	All day efficiency is considered
m efficiency of	Designed for 50-70% efficiency